PROJECT CHARTER

Smart Parking Platform · real-time parking discovery, reservation, payment and facility management

$oftware ¢orp for Bayou City Parking Group
Charter v1.0 · issued with SRS v7.0 · October 29, 2026
Status: **Approved at M1 (Oct 16, 2026)** · supersedes none

PROJECT MANAGER
Faiz Hajiani, $oftware ¢orp

EXECUTIVE SPONSOR
Bayou City Parking Group leadership (client)

PRODUCT OWNER
BCPG operations lead (accepts scope and milestones)

INTERNAL SPONSOR
$oftware ¢orp leadership (staffing, margin)

START · END
Sep 8, 2026 · Apr 5, 2027 (142 working days)

COST BASELINE (BAC)
$2,146,328 · total funding $2,253,644

CONTRACT
Milestone-based: 30% kickoff · 40% MVP · 30% final

GOVERNING DOCUMENT
SRS / Project Management Document v7.0 (GitHub)

PROBLEM STATEMENT

Drivers in Downtown, Midtown, the Medical Center and the Galleria circle for parking while BCPG's 14 garages and lots sit under-used. BCPG has no live occupancy view across facilities, reservations sold through third-party listings are not honored at the gate, pricing is static, and revenue reporting is done in spreadsheets.

BUSINESS CASE

One platform replaces the patchwork of listings, legacy pay stations and manual reports. Real-time inventory raises utilization and lets BCPG price by demand; reservations that open the gate cut disputes and attendant time; one vendor instead of two lowers operating cost. The contract was won on exactly these three commitments (SRS 2.2).

GOAL STATEMENT AND OBJECTIVES

- MVP (accounts, search, reservation, payment, gate entry, facility setup) accepted at two pilot facilities by **Feb 17, 2027**.
- All 14 facilities configured and Beta live for invited drivers by **Mar 3, 2027**.
- Public launch on App Store, Google Play and web by **Apr 5, 2027**.
- Service targets at launch: availability updates within 5 s (NFR-2), gate authorization under 1 s at p99 (NFR-3), 99.9% / 99.95% uptime (NFR-4), WCAG 2.1 AA (NFR-11).
- Deliver within the $2,146,328 baseline; CPI and SPI ≥ 0.90 every month.

SCOPE

In: iOS/Android apps and responsive web; real-time availability from gates, sensors and LPR; map and hand-off to navigation; reservation engine (hourly, daily, event, monthly); PCI-compliant payments and receipts; operator dashboard (rates, capacity, occupancy, reporting); notifications; role-based administration and audit; integrations with gate/LPR hardware, payment and mapping providers; open-data feed.
Out: on-street metered parking and enforcement; hardware manufacture or installation; valet; EV charging; public marketplace for non-client operators; loyalty and advertising; custom city permit integrations. Changes go through a formal change request (SRS 2.2, 14.8).

KEY DELIVERABLES

- Requirements, architecture, data model and UX designs (M1)
- Backend/API, mobile apps, web app, operator dashboard
- Payment, mapping and gate/LPR integrations; edge buffering service
- CI/CD pipeline, staging with production parity, delivery dashboard
- Automated test suite, security review, pilot and Beta reports
- Documentation, attendant and operator training, handover pack

MILESTONES

ID	Milestone	Date	Payment
M0	Kickoff / contract award · SRS v1.0	Sep 8, 2026	30%
M1	Requirements and design approved · build gate	Oct 16, 2026	—
M2	MVP accepted at two pilot facilities	Feb 17, 2027	40%
M3	Beta release · 14 facilities · security sign-off	Mar 3, 2027	—
M4	Public launch, training and handover	Apr 5, 2027	30%

PROJECT TEAM

20 people in six cross-functional teams plus the PM: Solutions Architect / Backend Lead (Daniel Okafor) and five backend engineers; Mobile Lead (Grace Kim) + 2; Web Lead (Nathan Wright), UX designer, frontend engineer; Mapping (Hiro Tanaka, Lucy Nguyen); Payments (Robert Stein, Chloe Dubois); QA Lead (Angela Morris) + 2; DevOps / Cloud (Samir Haddad). Roles, allocations and RACI in SRS 14.

KEY STAKEHOLDERS (SRS 15)

Manage closely: client product owner; team leads and architect. **Keep satisfied:** client executive sponsor; $oftware ¢orp leadership; course instructor; regulators; app-store review; gate/LPR vendors. **Keep informed:** development teams, facility managers, client IT, finance, attendants, drivers. **Monitor:** payment and mapping providers, competitors.

TOP RISKS (SRS 11)

- **T-1 / P-3** Gate and LPR vendor APIs differ from documentation; team lacks IoT depth → lab kits, two integration windows, vendor check-ins.
- **S-5** Client data or app-store approval arrives late → early store accounts, Beta before launch.
- **T-4** Faulty production release at a live facility → release readiness review, degraded mode (FR-29).
- **S-1** Hurricane closes the Houston office → remote-first, 18-day schedule cushion.

ASSUMPTIONS AND CONSTRAINTS

- Budget is not the constraint; the fixed academic timeline is (final document Dec 4, 2026; delivery Apr 5, 2027).
- Client runs on Microsoft 365 (confirmed at M1); Jira and GitHub are the systems of record.
- PCI scope stays with the payment provider; Texas and City of Houston ordinances and WCAG 2.1 AA apply.
- Vendors provide sandbox and firmware access before Nov 4, 2026.

PROJECT MANAGER AUTHORITY AND SUCCESS CRITERIA

The Project Manager is authorized to assign the resources in SRS Table 14.1, run the schedule and sprint cadence, approve contingency draws up to $25,000 per item, and sign releases to Staging. Scope changes, contingency above $25,000 and any Production release need the Client Product Owner's written approval; management reserve needs the Executive Sponsor. **Success** = every milestone accepted in writing by its target date, no Severity-1 defect open at launch, service targets met in the launch stress test, and the handover pack accepted by client IT.

Executive Sponsor, Bayou City Parking Group
Signature · Date

Leadership, $oftware ¢orp
Signature · Date

Faiz Hajiani, Project Manager
Signature · Date